

Body Science Checkpoint

I can use models and evidence to show what I know about skeletons, muscle pairs, nutrition, balance and food sources.

SKELETON

backbone evidence

MUSCLES

pull to move

NUTRITION

evidence, not judgement

[Open the original interactive lesson](#)

What success looks like

- I can show one secure idea from each checkpoint strand.
- I can connect at least two ideas using because.
- I can keep my first evidence and identify the strand to practise next.

Retrieve and reconnect

- Name one skeleton job and one bone example.
- Explain what one muscle does while its partner relaxes.
- Define vertebrate using backbone.
- Lead-in: state one supported claim from the sugar labels.

A pupil says, 'The bone pushes the muscle, so the arm bends.' Which idea should the checkpoint test?

Think first. Point, say, sign or write your choice. Give one reason.

- A. Keep it: bones push muscles to make them shorter.
- B. Test the pair: one muscle contracts and pulls while the other relaxes.
- C. Ignore the arm model and compare sugar labels instead.

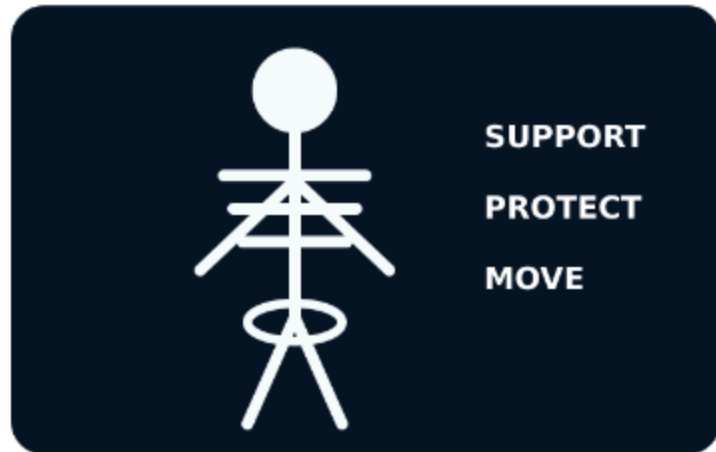
Look closely · what is the evidence?

SKELETON · bones give support, protection and places for movement

MUSCLE PAIR · one contracts and pulls; the partner relaxes

NUTRITION · food provides nutrients; animals obtain food by eating

Look closely · what is the evidence?



A simple skeleton with the words support, protect, move



Visual 2 · inspect the supplied evidence



Visual 3 · inspect the supplied evidence

Words we will use

- support — hold up and give shape
- contract — shorten and pull
- nutrition — getting what the body needs from food

Watch the model

- Checkpoint 1: use a skeleton model to match support, protection and movement to an example; classify one animal using backbone evidence.
- Checkpoint 2: operate the arm model. Name which muscle contracts, which relaxes and how the bone moves at the joint.
- Keep each pupil's first response. Feedback adds a repair beside it rather than erasing the evidence.

Watch the model



Arm model showing biceps and triceps

When the biceps contracts to bend the elbow, what should the triceps do in the simple model?

- A. Relax while the biceps pulls the forearm bone.
- B. Push the bone from behind.
- C. Contract fully in the same direction as the biceps.

Commit to an answer. Show the evidence you used. Listen, then revise if needed.

Build the explanation

- Checkpoint 3: match food evidence to body jobs such as energy, growth and repair; explain why variety matters.
- Checkpoint 4: sort foods with a stated rule and build a simple producer-to-consumer food-source chain.
- Connect W8A carefully: a sugar value is one piece of nutrition evidence, not a score for a food or person.

Find and repair the misconception

Checkpoint glitch: bones move by themselves and food with less sugar always makes every body healthier.

A. Keep both claims because they sound simple.

B. Repair both: muscles pull bones in pairs, and sugar is only one part of nutrition evidence.

C. Remove the first answers so nobody can see the misconceptions.

Four-Strand Body Science Checkpoint

Route each evidence card to the strand it checks. Then choose one strand to explain and one to revisit.

- Choose one evidence card.
- Place it with its strand.
- State or point to the science idea.
- Repair after feedback without deleting the first response.
- Choose the next strand.

Four-Strand Body Science Checkpoint

Use the numbered cards in your pupil pack. Take turns to choose, justify and check. You may direct a partner to move a card.

Choose a group · defend your placement

- SKELETON / BACKBONE
- MUSCLE PAIR
- NUTRITION / BALANCE
- FOOD SOURCE / CHAIN

Name one secure strand, one connection between strands and the exact next strand to revisit.

Your independent mission

Complete or direct four short checkpoint stations, preserve a piece of evidence from each and explain one connection between two strands.

Record: One observable response from each strand, one because connection, the pupil's chosen next strand and an adult witness note naming independence and prompts.

Choose your support and challenge

- Supported: Complete one two-choice item at each station, then choose a secure strand and a next strand.
- Standard: Complete all four stations, justify one response from each and connect two strands using because.
- Stretch: Explain two cross-strand links, repair one misconception and evaluate one model limitation.

Show what you know

Explain one connection between skeletons, muscles, nutrition or food sources using evidence.

Point to, read or explain the evidence you made. Choose one next step after the adult has listened.